

Optimizing Animal Shelter Outcomes

Predicting Adoption Likelihood to Guide Shelter Resource Allocation

August 2026

Executive Summary

Animal shelters operate under tight constraints on space, veterinary budget, and staff time, and every extra day an animal spends unadopted adds cost and risk. This project builds a classification model that predicts, at intake, whether a shelter animal is likely to be adopted, using a Kaggle dataset of 2,007 shelter animal records (species, breed, age, weight, size, color, vaccination and health status, adoption fee, time in shelter, and prior ownership). Three modeling approaches were compared: logistic regression, a decision tree, and a random forest, each tuned via 5-fold cross-validated grid search. The random forest performed best, correctly classifying 91.3% of test-set animals with an F1 score of 0.863, well ahead of a majority-class baseline (67.2% accuracy, F1 of 0). The two strongest, most actionable predictors of adoption were vaccination status and health condition: vaccinated animals were adopted at nearly four times the rate of unvaccinated animals (42.1% vs. 11.2%), and animals with a documented health condition were adopted at less than a quarter the rate of healthy animals (9.6% vs. 38.5%, a statistically significant difference, $p < 0.001$). These findings translate into three concrete recommendations for shelter operations, detailed in Section 7.

1. Problem Identification

1.1 The Problem

Shelters often lack data-driven insight into which animal characteristics most influence adoption likelihood, making it hard to know where to direct limited resources. Left unaddressed, animals who could benefit most from early intervention -- extra medical care, targeted marketing, or a spot at a themed adoption event -- may not be identified until they have already become a "long-stay" case.

1.2 Business Goal

The goal is a predictive classification model that flags high-risk, low-adoption-likelihood animals early in the intake process, so shelter staff can:

- Allocate resources -- prioritize high-need animals for specialized training or medical care.
- Target marketing -- focus photography and social media effort on animals predicted to have lower adoption probability.
- Optimize events -- design adoption events around the traits currently showing the weakest engagement.

1.3 Stakeholders

- Shelter operations managers -- kennel capacity and staffing decisions.
- Veterinary staff -- which health conditions present the biggest barrier to adoption.
- Marketing/outreach teams -- justification for campaign and platform spend.
- The animals -- the ultimate beneficiaries of a shorter length of stay and a better-matched home.

1.4 Scope

In scope: exploratory analysis of animal traits, feature engineering, and training/evaluating classification models. Out of scope: real-time deployment into shelter software, and computer-vision analysis of animal photos -- both noted as candidates for future work.

2. Data Overview

The dataset ("Predict Pet Adoption Status," Kaggle) contains 2,007 shelter animal records with 13 raw fields: a record identifier; four species types (bird, cat, dog, rabbit) across seven breeds; five colors; three size classes; age in months; weight in kilograms; vaccination status, health condition, and prior-ownership flags (binary); adoption fee; time in shelter (days); and the binary target, AdoptionLikelihood (0 = not likely, 1 = likely). Roughly one third of animals in

the dataset were adopted (32.8%), making this a moderately imbalanced classification problem -- a constraint that shaped both the choice of evaluation metric (Section 3.4) and the model comparison (Section 4.1).

The dataset has some real limitations worth naming up front: it contains no behavioral or temperament data, which likely also influences adoption outcomes but isn't captured here, and no image data, so any future computer-vision extension (e.g., photo quality) is out of scope for this phase.

3. Methodology

3.1 Data Wrangling

Initial checks confirmed the dataset had no missing values requiring imputation. Descriptive review of the numeric ranges surfaced a few possible data-collection artifacts worth flagging rather than silently trusting: ages range from 1 to 179 months (about 15 years, plausible for a senior dog, but 1 month may be a data floor rather than a true minimum), weight spans roughly 1 to 30 kg (a wide but plausible range across birds, rabbits, cats, and dogs), and time in shelter tops out at 89 days, which may reflect a genuine maximum or a reporting cutoff (e.g., a quarterly reset). None of these required correction, but they are documented as limitations on how far the model's conclusions should be extrapolated.

3.2 Exploratory Data Analysis

Univariate analysis showed the four continuous features (age, weight, time in shelter, adoption fee) were all reasonably close to symmetric, with no skew severe enough to warrant a log transform. Bivariate analysis against the target surfaced three patterns that carried through to the final model and the recommendations in Section 7:

- Vaccination status: 42.1% adoption rate for vaccinated animals vs. 11.2% for unvaccinated.
- Health condition: 38.5% adoption rate for healthy animals vs. 9.6% for animals with a documented condition -- confirmed statistically significant with a chi-square test of independence ($\chi^2 = 118.24$, $p < 0.001$).
- Species: dogs had a notably higher adoption rate (46.4%) than birds (30.2%), cats (28.7%), or rabbits (25.4%).

Prior ownership showed almost no relationship to adoption outcome (33.5% vs. 31.2%), and a Pearson correlation check across the numeric features found no concerning multicollinearity, so no upfront dimensionality reduction (e.g., PCA) was necessary. Category cardinality was also checked and confirmed low across all categorical fields (2-7 unique values each), so no category consolidation was needed before encoding.

3.3 Preprocessing

The record identifier (PetID) was dropped as non-predictive. A crosstab of Breed against PetType showed every breed maps to exactly one pet type in this dataset (e.g., every parakeet is a bird), so PetType was dropped as fully redundant with the more granular Breed field. Size and an engineered LifeStage feature (grouping age into infant/young adult/adult/senior bands) were ordinal-encoded to preserve their natural ranking; the remaining nominal categoricals (Breed, Color) were one-hot encoded with the first category dropped to avoid perfect collinearity. The data was split 80/20 into training and test sets, stratified on the target to preserve the 32.8% positive rate in both splits, and the four continuous numeric features were standardized with a scaler fit only on the training data to avoid leakage.

3.4 Modeling

Three classification algorithms were trained and compared, plus a majority-class baseline to establish a performance floor: logistic regression (an interpretable linear baseline), a decision tree, and a random forest. Each was tuned via 5-fold cross-validated grid search. For the two tree-based models, the split criterion (Gini impurity vs. entropy) was included directly as a tuning parameter rather than assumed in advance, so the search itself determined which criterion fit the data better. Because the target is imbalanced, F1 score -- rather than raw accuracy -- was used as the grid search scoring metric and the primary basis for model selection, since accuracy alone can look deceptively good while a model fails to identify any true adoptions (as the baseline demonstrates in Section 5).

4. Results

4.1 Model Comparison

All three tuned models comfortably outperformed the majority-class baseline. Results on the held-out test set:

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.913	0.894	0.833	0.863	0.895
Decision Tree	0.910	0.881	0.841	0.860	0.887
Logistic Regression	0.699	0.531	0.720	0.611	0.796
Baseline (Majority Class)	0.672	0.000	0.000	0.000	0.500

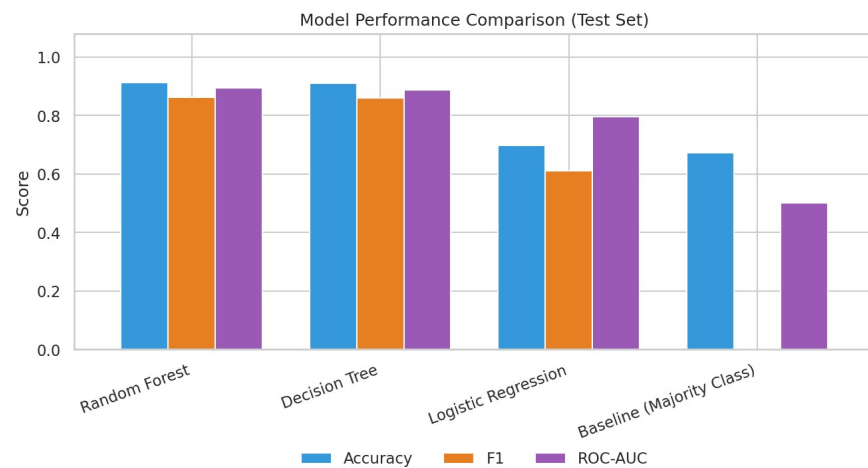


Figure 1. Test-set performance across all four models. The tree-based models clearly separate from the baseline and the linear model.

4.2 Final Model Selection

Selected model: Random Forest Classifier. It achieved the highest F1 score (0.863) and accuracy (91.3%) of the models tested, tuned to 100 trees, entropy split criterion, unrestricted max depth, and balanced class weighting via grid search. The decision tree was a close second (F1 0.860) with the simpler, more interpretable single-tree structure and a gini split criterion -- a reasonable alternative if interpretability or training cost became a priority. Logistic regression, while far more interpretable through its coefficients, could not capture the non-linear relationships in the data as well (F1 0.611), confirming this is a case where the added complexity of an ensemble model is worth the trade-off against a simpler linear one.

4.3 What Drives the Predictions

The random forest's feature importances (Figure 2) point to Size, AgeMonths, and Vaccinated as the three strongest predictors, followed closely by AdoptionFee, WeightKg, and TimeInShelterDays. HealthCondition and the Labrador breed indicator also contribute meaningfully. This is broadly consistent with the logistic regression coefficients, where Vaccinated (+2.21) and the Labrador breed indicator (+2.29) were the strongest positive predictors of adoption, and HealthCondition (-2.35) and LifeStage (-1.19) were the strongest negative ones -- two independent modeling approaches converging on the same handful of features builds confidence that these relationships are real rather than an artifact of one algorithm.

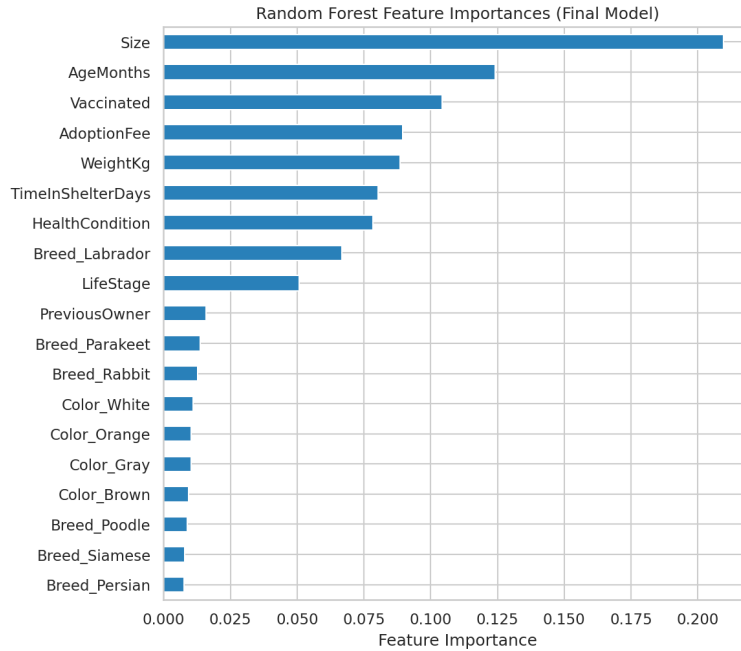


Figure 2. Random forest feature importances for the final model.

4.4 Predictions vs. Actual Outcomes

On the 402-animal test set, the final model correctly classified 367 animals (257 true negatives, 110 true positives) and misclassified 35 (13 false positives, 22 false negatives).

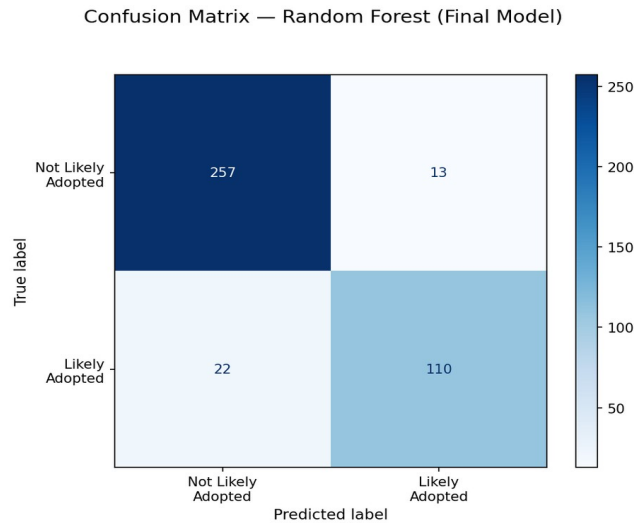


Figure 3. Confusion matrix for the final model on the test set.

Precision and recall are both strong for the "not likely adopted" class (0.92 / 0.95) and solid for the "likely adopted" class (0.89 / 0.83), which matters for the intended use case -- correctly identifying which animals to prioritize matters more than a marginal accuracy gain. One nuance worth flagging: among the 22 false negatives (animals the model predicted would not be adopted but were), 50% had a documented health condition, compared to 22.6% across the test set overall. In other words, the model may somewhat over-weight health condition as a negative signal, occasionally underestimating the adoption chances of animals with health conditions who are, in fact, adopted. This is worth monitoring in a production setting so genuinely adoptable animals with health conditions aren't systematically deprioritized for marketing or events (see Section 6).

5. Key Findings

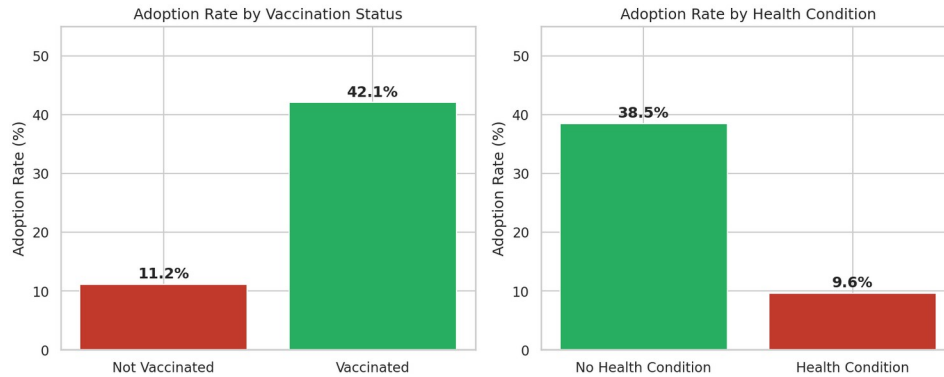


Figure 4. Adoption rate by vaccination status (left) and health condition (right).

Two operationally controllable factors -- vaccination and health condition -- show the largest, most statistically robust differences in adoption rate of any features examined, and both are actionable by shelter staff rather than fixed traits of the animal. Species also matters: dogs adopt at meaningfully higher rates than the other three species in this dataset, which may reflect real demand differences worth accounting for in event planning and marketing allocation.

6. Recommendations

- 1. Vaccinate at intake, not on a delayed schedule.** Vaccinated animals in this dataset were adopted at nearly four times the rate of unvaccinated ones (42.1% vs. 11.2%). If vaccination is currently scheduled later in an animal's stay, moving it as early as clinically possible is a low-cost, high-leverage change directly within shelter operations' control.
- 2. Fast-track veterinary care and marketing support for animals with health conditions.** The gap here is severe and statistically confirmed (9.6% vs. 38.5% adoption rate, $p < 0.001$). Rather than treating a health condition as a reason an animal will simply wait longer, shelters could pair expedited treatment with dedicated photography/marketing for these animals specifically -- directly serving both the veterinary and marketing stakeholder goals identified in Section 1.3.
- 3. Use model-predicted adoption likelihood to flag at-risk animals within 30 days of intake.** Scoring animals at intake and routing low-probability predictions to staff for extra attention operationalizes the original 70%-of-high-risk-animals-identified-in-30-days target from the project proposal. Because dogs already outperform other species by a wide margin, this is also a natural opportunity to design species-specific adoption events for birds, cats, and rabbits rather than one-size-fits-all campaigns.

7. Limitations & Further Research

- No behavioral or temperament data is available, which likely affects adoption outcomes but isn't captured in this dataset.
- No image data -- photo quality is a plausible driver of adoption interest and was explicitly out of scope for this phase.
- This is a Kaggle dataset rather than a live shelter feed; relationships should be re-validated against real operational data before any recommendation is acted on.
- The model shows a mild tendency to under-predict adoption for animals with health conditions who are ultimately adopted (Section 4.4) -- worth monitoring rather than trusting blindly in production.
- Next steps: test gradient boosting (e.g., XGBoost) against the random forest; treat time-to-adoption as a survival-analysis problem rather than a binary outcome; and, once deployed, track the model's real-world 30-day identification rate against the 70% target set in the original proposal.

8. Conclusion

A tuned random forest classifier predicts shelter animal adoption likelihood with 91.3% test-set accuracy and an F1 score of 0.863, well ahead of both a naive baseline and a linear model. Vaccination status and health condition emerged as the two most actionable, statistically significant drivers of adoption outcome, translating directly into concrete operational recommendations. With the caveats noted above -- particularly the need to validate these patterns against live shelter data before deployment -- this model and the findings behind it give shelter operations, veterinary, and marketing staff a data-driven starting point for prioritizing the animals who need help the most.